Behavior Analysis: Research and Practice
2017, Vol. 17, No. 1, 92–97

2372-9414/17/$12.00 http://dx.doi.org/10.1037/bar0000055

RESEARCH REPORT

Evaluating Emergent Naming Relations Through Representational Drawing in Individuals With Developmental Disabilities Using the PEAK-E Curriculum

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Naming has been described behavior analytically as the ability to engage in both speaker and listener behavior as they relate to stimuli in the real world, where speaker behavior is often demonstrated in the form of tacting and listener behavior in the form of selection-based responding. The present study sought to evaluate *representational drawing* as an alternative to selection-based listener responding where the listener behavior is unprompted. A multiple-baseline with an embedded multiple probe design was conducted across 3 individuals with developmental disabilities, and tact/drawing stimuli included unfamiliar animal blends. The results suggested that, following tact training, all of the participants demonstrated the untrained emergence of representational drawing of the blended animal pictures. The procedures and stimuli in the present study were taken from the PEAK-E curriculum, allowing for potentially easier replication by practitioners and researchers.

Keywords: autism, disabilities, naming theory, representational drawing

Naming is defined in the behavior analytic literature as the ability to engage in both a speaker response and listener response as they relate to stimuli in the real world (Horne & Lowe, 1996), and has been described as fundamental in the typical development of symbolic language (Greer & Keohane, 2005). Horne and Lowe (1996) suggest that in typical language development, children progress from early listener behavior to echoic and subsequently tacting behaviors, the combination of which comprise the behavior of symbolic naming. (For a review of the role of speaker and listener behavior in the development of complex language, see Schlinger, 2008). This co-occurring set of behaviors may be pivotal in the development of more complex topographies of language such as those shown in stimulus equivalence paradigms (Horne & Lowe, 1996). Several studies have since evaluated how naming emerges in young children (e.g., Miguel, Petursdottir, Carr, & Michael, 2008), and how training approaches grounded in naming theory have resulted in the subsequent emergence of untrained verbal skills in individuals with autism and related disabilities who are likely to experience language deficits (e.g., Kobari-Wright & Miguel, 2014; Miguel & Kobari-Wright, 2013). For example, Greer, Stolfi, Chavez-Brown, and Rivera-Valdes (2005) showed how multiple exemplar training of listener (i.e., point-to) and speaker (i.e., pure and impure tacts) relations resulted in the ability of participants with developmental delays to derive untrained speaker responses following direct training of listener respond-

This article was published Online First September 15, 2016.

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Mark R. Dixon receives small royalties from the sales of the PEAK curriculum.

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ing. In a similar study, Delfs, Conine, Frampton, Shillingsburg, and Robinson (2014) compared the efficacy of listener and speaker training on the emergence of the untrained speaker and listener responses in children with autism, suggesting that emergent speaker or listener skills were apparent given both training modalities.

As suggested by Greer and Keohane (2005), several response topographies can be emitted that allow for a demonstration of emergent naming, including "match-to-sample, point-to, pure tact, and impure tact responses" (p. 124). Over the past few decades, cognitively oriented researchers have evaluated a phenomenon called *representational drawing*, which may provide an additional method for evaluating *naming*. Representational drawing is the ability to draw objects or pictures that a person has observed (see Cox, 2005). Like naming, representational drawing appears to be a milestone in typical development (Cox, 2005) and is reflected in several items in the common core standards guiding typical educational practice in the United States (e.g., CCSS.ELA-Literacy.W.K.2; National Governors Association Center for Best Practices and Council of Chief State School Officers, 2010). In addition, representational drawing appears to be a deficit experienced by individuals with autism (Ford & Rees, 2008). Although representational drawing may initially involve drawing objects in the immediate environment (i.e., see—draw), later representational drawing occurs when only the name of the object is provided in the absence of the physical object to which the name refers (i.e., hear—draw). Whereas the speaker response of naming involves tacting an object (i.e., see—name), representational drawing may allow for a demonstration of the listener response (i.e., hear—draw), where the listener response is unprompted, unlike "match-to-sample" and "point-to" arrangements that provide a stimulus array from which to select the stimulus (i.e., hear—point). Representational drawing may demonstrate *naming* when a participant is able to, following learning to tact the object, draw the object when provided the object name, allowing for a potentially stronger demonstration of a *naming* relation by eliminating the need for a stimulus array and therefore reducing the probability of correct responding due to visual prompt or chance alone, having especial utility for individuals who demonstrate sophisticated verbal repertoires.

The purpose of the present study was therefore to evaluate representational drawing as an alternative to selection-based listener responding in an evaluation of naming by individuals with disabilities. Participants were first taught to tact blends of common animals, and test probes of untrained representational drawing were conducted throughout the study. Animal blends were targeted in the current study to allow for the use of stimuli with which participants were unlikely to have a specific learning history, as well as to provide an opportunity for creative responding and problem solving that has not been directly learned given the difficulties associated with producing flexible and creative behavior associated with autism and related disabilities. The training procedures and stimuli were taken directly from the Promoting the Emergence of Advanced Knowledge Equivalence Module (PEAK-E; Dixon, 2015), a curriculum designed to develop language skills through stimulus equivalence procedures. The procedures described in PEAK-E were used to aid in replication both clinically and in research.

Method

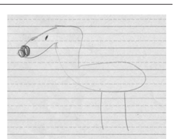
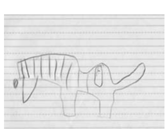
Participants, Setting, and Materials

Three males, Jerome (8 years old), Ronald (10 years old), and William (7 years old), participated in the study. Jerome was diagnosed with autism; Ronald was diagnosed with autism and attention deficit and hyperactivity disorder; and William was diagnosed with an emotional and developmental disability. All of the participants received the maximum possible score on the Verbal Behavior Milestones Assessments and Placement Program (VB-MAPP). Prior to the study, each of the participants were able to demonstrate reflexive and symmetrical arbitrary relational responding, as well as basic and intermediate transitive relational responding on the PEAK-E Preassessment (PEAK-E-PA). Prior to the study, participants were able to tact the unblended animals used in the study and draw several familiar animals that were not used in the study. All of the training and testing sessions were conducted either in the participants' home classrooms or in a separate room at the school. The authors of the study conducted all sessions and were familiar with the students prior to the study. Sessions lasted between 20 min and 2 hr each day, and were conducted up-to once per day over three weeks. Individualized skills unrelated to the study were also targeted through the PEAK curriculum during the instructional sessions, accounting for much of the differential lengths of each session. The materials included three pictures of blended animals (see Table 1), a pencil, and several 8- × 11-in. white sheets of paper.

Dependent Variable and Interobserver Agreement

Two behaviors were evaluated in the present study, where tacting served as the speaker response and drawing served as the listener response. The "correctness" of participant responses were reported both as a percentage of correct responses within a trial block (i.e., the number of correct independent trials divided by the total number of trials, multiplied by 100), as well as a calculated PEAK score within five-trial blocks, where scores for each trial ranged from 0 to 10 based on the prompt level required for correct responding. PEAK scores were reported to improve replication of the procedures by individuals implementing the PEAK curriculum and because of the use of PEAK scores in prior literature (e.g., McKeel, Rowsey, Belisle, Dixon, & Szekely, 2015). The rating for each of the five trials was summed and multiplied by 2 to arrive at a PEAK score between 0 and 100 for each trial block. Scores were multiplied by two to remain consistent with other studies irrespective of the number of trials within a trial-block. Interobserver agreement (IOA) was calculated by comparing the scoring of two independent observers during both training and probe sessions using the obtained PEAK scores (IOA on a PEAK score for a given trial would also indicate IOA for percent correct). The total number of agreements between the two observers was divided by the sum of the number of agreements and the number of disagreements, multiplied by 100. IOA was collected for both target behaviors during a combined 34% of the trial blocks, and IOA was calculated at 96%.

Table 1
Stimuli Used in the Present Study and Exemplars of Participant Representational Drawings

Animal blend	Picture stimulus
Ostrich/Horse	
Spider/Owl	
Elephant/Zebra	

Design and Procedure

A multiple baseline (AB) with an embedded multiple probe design across participants was used in the present study, where the A phase was a baseline phase and the B phase was a Tact Training phase.

Baseline. In baseline, test probes were conducted for both the tacting and drawing responses to evaluate the participants' ability to demonstrate both of these skills prior to tact training. For tact probes, the participants were shown a picture of an animal blend (e.g., an ostrich/horse) and asked, "What is this called?" where a response was considered correct if the participant provided a tact that contained at least a syllable from each of the animal names included in the blended picture (e.g., calling a blend of an ostrich and a horse an *ostrihorse*). For representational drawing probes, the participants were provided a piece of paper and a pencil, and asked to "Draw a(n) *x*," where *x* was a blended animal name (e.g., "Draw an ostrihorse"). A response was considered correct if the participant drawing contained at least a single feature from each of the animals in the blend that was specific to the unblended animal (e.g., a horse tail and a beak), and these feature could not be a shared feature across the two animals (e.g., both a zebra and elephant have four legs). For IOA, observers had to agree on the score assigned to each of the drawings. The participants were exposed to each of the three picture stimuli during each trial block, where two of the three pictures were presented twice within each block. The order of the stimuli were selected by the experimenter prior to the trial by writing an unordered sequence of numbers 1, 2, or 3, each number corresponding to a stimulus class. The participants were given up to 2 min to complete each drawing or until the drawing was completed as indicated by the participant removing the tip of the pencil from the paper for 10 s or greater. During this phase, no differential consequences were delivered for correct nor incorrect responses, and no prompts were used.

Tact training and drawing test probes. In the present study, tact training was exclusively conducted and test probes of representational drawings were periodically implemented to test for the emergence of untrained representational drawing. Tact training involved the experimenter holding up a picture of the blended animal and asking, "What is this called?" where a response was considered correct if the participant provided a tact that contained at least a syllable from each of the animal names included in the blended picture. Participants were provided verbal praise following correct responses, as this was identified as a preferred stimulus by the classroom teacher. If the participant did not give an appropriate name within 10 s, the instructor used a least-to-most intrusive prompting sequence. A first prompt involved providing the participant with a letter-sound of one of the animals in the blend (e.g., *s* for spider), where a subsequent correct response resulted in a PEAK score of 8. A second prompt involved providing the participant with a syllable-sound of one of the animals in the blend (e.g., *spid* for spider), resulting in a PEAK score of 4; and a final prompt involved providing the participant with the complete animal blend (e.g., *spidowl* for spider/owl), resulting in a PEAK score of 2. Test probes for representational drawing were identical to the baseline phase resulting in a score of 0 or 10. Participants were provided a 2-min break or greater prior to the onset of test probe trial blocks to ensure a delay between seeing the picture and the representation drawing task.

Results and Discussion

Figure 1 displays participant scores in the baseline and tact training phases. In baseline, Jerome had a mean percent correct and PEAK score of 0 for both tacting and representational drawing. During the tact training phase, there was variability for both types of responses, resulting in a mean percent correct score of 60 and a PEAK score of 74.9 for tacting, and reaching the mastery criterion of 100 for both percent correct and PEAK score on the final trial block. Test probes of representational drawing suggest that the mean percent correct and PEAK score increased to 60, and a score of 100 was achieved for the final three trial blocks. For Ronald, all baseline percent correct and PEAK scores for both tacting and representational drawing were also 0. During the training phase, tact training resulted in a mean percent correct of 88 and a PEAK score of 91.3. Following two training trial blocks, all trial blocks resulted in a percent correct and PEAK score of 100. For representational drawing, Ronald had a mean percent correct and PEAK score of 40, and achieved a percent correct and PEAK score of 100 consistently following the second test probe. William had a mean baseline percent correct and PEAK score of 10 for both tacting and representational drawing in the baseline phase. During tact training, the first trial-block resulted in a percent correct of 60 and a PEAK score of 90, with all subsequent trial blocks reaching 100. Test probes of representational drawing resulted in an initial percent correct and PEAK score of 60, with all subsequent scores consistently at 100.

The results of the present study support the use of representational drawing in the evaluation of *naming,* and add to a growing body of literature supporting the development of naming repertoires in individuals with- and without-disabilities (e.g., Kobari-Wright & Miguel, 2014; Miguel & Kobari-Wright, 2013; Miguel et al., 2008). Following tact training, all of the participants demonstrated untrained representational drawing of the target animal blends, a skill that they were unable to demonstrate prior to tact training. Given the preexisting verbal repertoire of the participants, the use of *drawing* may strengthen a demonstration of naming by eliminating the need for a stimulus array. A limitation in the present study was that the re-

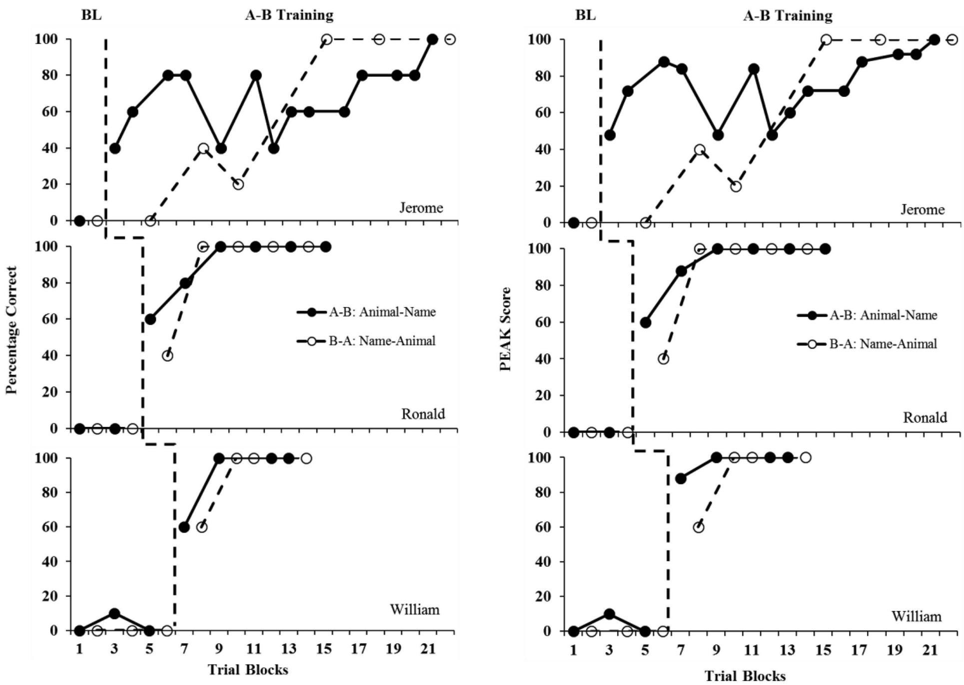


Figure 1. Participant percent correct (left panel) and Promoting the Emergence of Advanced Knowledge (PEAK) scores (right panel) for tacting and representational drawing. Closed circles represent tact trial blocks and open circles represent representational drawing trial blocks.

sults obtained may be unlikely in application with individuals who are not already able to tact and representationally draw. Similar limitations are present in the results obtained in other studies on naming (e.g., Delfs et al., 2014) in that emergent listener selection responses are unlikely if participants are not already able to select stimuli from an array and emergent tact responses are unlikely if participants are unable to tact. A major strength of the study conducted by Greer et al. (2005) was that multiple exemplar training was used to teach both tacting and selecting given these skills were not already present, and a similar arrangement may be considered for replication of the present study in application with individuals with greater language deficits. A second limitation in the present study is that several aspects of naming, such as echoic responding (Horne & Lowe, 1996) were not evaluated and therefore the process of naming is an assumption rather than an empirically evaluated phenomenon in the current investigation. This limitation may also be addressed in future research where drawing is used in evaluating derived listener responding. In summary, the results of the present study provide a first demonstration of the emergence of representational drawing as listener behavior following tact training. The data add to a body of literature supporting the role of naming in the emergence of untrained language skills, and provide a procedure for testing naming relations where the listener response is unprompted by a stimulus array as in "match-to-sample" and "point-to" procedures. The results further support the use of the standardized procedures in the PEAK-E curriculum in application with individuals with autism and related disabilities, providing researchers and clinicians with a prescribed set of instructions for training relations and testing for derived relational responding.

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Received April 18, 2016
Revision received July 11, 2016
Accepted July 19, 2016 ■